Luna Studio: Project Review vs. the Lunar Landing Site Navigator Challenge

Review of the uploaded codebase against the NASA Space Apps 2026 "Lunar Landing Site Navigator" challenge

Summary

Overall it's a polished, well-structured app with a strong visual identity, and the core ephemeris, timeline, and horizon-chart work is genuinely good. But several features are disconnected, a few numbers are made up, and the clearest gaps against the challenge are in scenario comparison and date handling.

Scope note: nothing in `utils/`, `data/`, `cesium/`, `workers/`, or `services/` was visible, so the ephemeris, libration, horizon table, and CLPS data are unverified. Some points below are from memory and are flagged.

1. What's broken

Features that don't work or can't be reached

- **The measurement tool is dead.** `handleToggleMeasure` and `M` call `mapRef.current?.activateMeasurement()`, but `mapRef` is never attached. The 2D view is `SouthPoleTopoMap` (Leaflet), which has no measurement API, so the toolbar opens and never gets points.
- **`MissionExplorerModal` is unreachable.** `UnifiedToolbar` destructures `onOpenMissionExplorer` but never uses it, and there's no shortcut. `onOpenComparison` isn't even destructured, so Comparison only opens from the Compare button on a selected Moon feature.
- **2D mode silently shows the wrong place for any site outside 80–90°S.** `flyTo` clamps latitude to -80, and `selectFeature` only knows the 12 hard-coded polar landmarks. Selecting Apollo 11 or Blue Ghost in 2D does nothing or lands you at -80°. The DEM only covers the polar cap, so the UI should say that and offer a switch to 3D.
- **The global Escape handler is stale.** The effect dependency list omits `isDteOpen`, `isSiteAnalysisOpen`, and `isWeatherOpen`, so Escape closing those modals works only by accident.

- **"Toggle all symbols" uses the wrong IDs.** It lists `lunar-landing-sites`, while `LayerToggle` uses `apollo-landing-sites` and `robotic-landing-sites`.
- **Layer toggles do nothing in 2D**, since `mapRef` is null.

Modals that ignore the chosen date

- `DteCalculationModal`, `SiteAnalysisModal`, and `DteSignalChart` read `celestialTimeStore.get().ms` once and never subscribe. If you scrub the timeline or change the date, they keep showing the old time. `DteSignalChart` builds its 30 days from the time at memo creation. This directly undermines "date flexibility."

Layout and display

- `FeaturePanel` (`bottom-[104px]`, right) sits on top of `MapControls` (`bottom-[110px]`, right), so selecting a site hides the zoom controls.
- `CelestialPanel` and `LayerPanel` both anchor at `top-14 right-3`. Layers closes Celestial when it opens, but pressing `E` doesn't close Layers.
- `DteCalculationModal` prints `station.latitude.toFixed(2)°N, …°E` for every station. Canberra shows "-35.40°N" and Goldstone shows "-116.89°E".
- The timeline uses 7–9 px text. That's too small for the "accessible to the public" goal.

2. Numbers that look real but aren't

The tool's job is to make decisions trustworthy, so these matter most.

1. **Custom coordinates have no terrain horizon.** `horizonAt(HORIZON_TABLE, id, az)` returns nothing for unknown IDs and everything treats it as 0°. A dropped pin at the pole gets a flat horizon with no warning, which is the opposite of the south-pole problem. The three catalogs (`LUNAR_FEATURES` / `LANDING_SITES`, `CLPS_MISSIONS`, `SOUTH_POLE_LANDMARKS`) also use different IDs, so a site picked in 2D can lose its horizon data.
2. **The landing verdict is a single-instant score.** It starts at 100 and subtracts fixed penalties based on whether it's day, night, or PSR right now, plus Earth line-of-sight to one station. A site can read "OPTIMAL LANDING SITE" at noon and fail a 14-day night. `sweepIllumination` and `sweepEarthLink` already exist, so the verdict should use sunlit %, longest dark period, and DTE window % over the chosen mission window.

3. **The signal-strength chart is a fictional link budget.** Transmit power, antenna gains, and a -135 dBm threshold are magic numbers, with arbitrary -12 dB and -160 dBm penalties by status. The comment says FSPL is "~216 to 218 dB," but 8.4 GHz at 384,400 km is about 222.7 dB. Either label it illustrative or remove it.
4. **The probe fallback invents terrain.** `elevationM ?? -1500` turns a failed DEM lookup into a plausible elevation.
5. **The DSN corridor on the 2D map is a hard-coded polyline.** It's a static drawing from the pole through Malapert, not a computed link.
6. **PSRs are circles of a guessed radius.** They should come from a real PSR dataset (LOLA/ShadowCam polygons) or be labeled "approximate".
7. **The fake elevation heuristic** in `TopographicalControlBar` uses `sin` / `cos` and displays "Cursor: +x m." It's only in the orphaned OpenLayers map, but delete it so it never ships.
8. **The loading bar is fixed at 88%**, and `PlanetViewport` force-hides the overlay after 2.4 s even if tiles aren't ready.
9. **Mission status labels are inaccurate.** "completed" shows as "Landed & Operating" and "failed" as "Suborbital / Terminated," which doesn't describe a lander like Peregrine. Also check the CLPS statuses against NASA's current page (`clpsMissions.js` was not seen).
10. **Some landmark coordinates look off.** From memory, Shackleton is about 89.9°S, 129.8°E (the code has -89.65) and Faustini about 87.3°S, 77°E (the code has 84.3°E). Check every row against the IAU Gazetteer, since the code comment claims they're "verified."

3. Against the challenge objectives

Objective	Status
Curated sites + custom coordinates	Good (search, pin drop, comparison inputs)
Side-by-side comparison	Partial: only 2 sites, no charts, no same-site-different-date mode
Date/time flexibility	Weak: date picker only in the 3D Celestial panel and the comparison modal, no date in 2D, modals ignore it

Objective	Status
Sun/Earth vs. horizon	Strong (`HorizonViewChart`) but only on custom sites if a horizon exists
Time to sunset/Earthset	Present, but buried inside the Site Analysis modal
Power/comms metrics	Sun elevation and sunlit % exist, with no power estimate or list of access windows
Multiple visualizations	Strong (3D, 2D, sky plot, timeline)
Temporal dynamics	Good timeline; the 2D map doesn't animate with time
Skill-level accessibility	Weak: no explainer or beginner mode, "About" is a dead button

4. What more you can do, highest impact first

1. **Compute horizons for any point from your own DEM.** `lola_sp_80s_int16.bin` and a worker already ship, so ray-march 360° azimuths in the worker for any clicked point. This fixes the missing-horizon problem and makes custom sites accurate.
2. **A launch-window heatmap.** Build a grid of days × sites, colored by sunlit % or DTE %, so a planner can see at a glance which dates work if the launch slips. This is the strongest answer to the challenge's schedule-change scenario.
3. **Better comparison.** Allow 3–4 sites, add "same site, date A vs. date B," overlay timelines, and rank sites in a table.
4. **Time-driven lighting on the 2D map.** Make the hillshade follow the Sun's azimuth/elevation at the chosen time, so scrubbing the timeline visibly moves the shadows. This is the most striking demo moment.
5. **Make the verdict honest.** Use window-based metrics, show the inputs, and add a power estimate (relative output on a horizontal and a vertical panel, longest night).
6. **Default to "any DSN complex."** The timeline only shows communication when both a Moon and an Earth feature are selected. Default to Earth-center or the union of Goldstone, Madrid, and Canberra, and list access windows with durations.
7. **Share and export.** Put site, date, and view in a URL hash, and add PNG/CSV export and a one-page "site card."
8. **A beginner mode.** Add a guided tour, a glossary, and an "Explain this chart" toggle.

9. **A validation page.** Compare sun and Earth elevations to JPL Horizons for a few site-times, and state whether libration is included. Judges will trust it more.
10. **A scale bar and a unified catalog on the 2D map.** The stereographic scale is close to 1 near the pole, so a scale bar is easy. Merge all the catalogs into one with stable IDs.

5. Cleanup

- **Dead code:** `LunarMap`, `SearchBox`, `SearchResults`, `ScaleControl`, `TopographicalControlBar`, `LunarWeatherWidget`, and `LunaBrandLogo` are unused. App still imports LunarMap, SearchBox, and LunaBrandLogo, `main.jsx` still imports `ol/ol.css`, and OpenLayers is no longer rendered. Removing it cuts bundle weight.
- **Palette drift:** `ErrorBoundary`, `LayerPanel`, `LayerToggle`, `MapControls`, `MeasurementToolbar/Result`, `ToastFeedback`, `CelestialPanel`, `LunarSunlightTimeline`, `MissionExplorerModal`, `SouthPoleLegendHUD`, `HorizonViewChart`, and `DteSignalChart` all still use slate/cyan.
- **Dialog accessibility:** modals lack `role="dialog"`, `aria-modal`, and focus trapping. Color-only status coding needs a second cue for color-blind users. Duplicate SVG IDs (`earth-globe`, `horizon-plot-clip`) will clash if two charts mount.
- **Unused imports:** `CompactClpsModal`, `DteCalculationModal`, and `LunarWeatherModal` import several unused icons, and `getMissionImage` / `getMissionAgency` are copied into two files.
- **Smaller issues:** `navigator.clipboard.writeText` has no `.catch`, `RealisticMoon` depends on CDN textures (cache them locally for demo day), and the title says "Lunar Explorer" while the brand says "Luna Studio."

6. Suggested next step

Fix the dead and stale items first (measurement, Mission Explorer/Comparison entry points, the 2D clamp, time subscriptions, panel overlap, the station sign formatting), then compute horizons from the DEM. Uploading `utils/`, `data/`, and `workers/` would allow a check of the ephemeris and horizon pipeline and the catalog data.

7. What's left, in priority order

The work splits into four tiers: make it trustworthy, finish the core loop, add what no other tool has, then polish. Tick items off as they land.

Tier 0: make it trustworthy (before any demo)

- [ ] Wire up or remove the measurement tool (Leaflet has no measure API yet)
- [ ] Add Mission Explorer and Compare to the toolbar
- [ ] Handle sites outside 80–90°S in 2D: show a message and offer 3D instead of clamping to -80°
- [ ] Subscribe the DTE modal, Site Analysis modal, and DTE chart to the shared clock
- [ ] Compute a horizon profile for any point, and warn when none exists
- [ ] Replace the single-instant verdict with window-based metrics
- [ ] Label or remove the link-budget chart, the -1500 m probe fallback, the 88% loading bar, and the static DSN line
- [ ] Fix panel overlap, station lat/lon signs, and the Escape handler
- [ ] Verify landmark coordinates and CLPS mission statuses against official sources

Tier 1: finish the core loop

- [ ] One global date/time control that works in both 2D and 3D
- [ ] Sunset and Earthset countdown on the feature panel, not only inside a modal
- [ ] Compare 3–4 sites, and the same site on different dates
- [ ] A list of Earth-access and sunlight windows with start, end, and duration
- [ ] A permalink that stores site, date, and view

Tier 2: differentiators (section 8)

- [ ] Launch-window heatmap and Plan B finder
- [ ] Time-driven lighting on the 2D polar map
- [ ] Lander profile presets
- [ ] Stand-there panorama
- [ ] Mission replay and a validation page

Tier 3: polish

- [ ] Beginner mode, glossary, and multiple languages
- [ ] CSV, calendar (.ics), and PDF export
- [ ] Energy and battery budget view
- [ ] Offline-safe demo (bundled textures, cached tiles)
- [ ] Palette unification, accessibility pass, and dead-code removal (section 5)

8. What would make it unique

Quickmap, Moon Trek, and JPL Horizons answer "what does the data say?" This app can own a different question: "will my mission work, and what do I do if it slips?" Everything below serves that question.

1. **Slip Simulator and Plan B finder.** Set a primary site and date, then drag a slider to slip the launch by hours, days, or a lunar month. The viability numbers update live. A "Find Plan B" button then searches later dates at the same site and nearby sites, and ranks them against the user's constraints. This is the challenge's schedule-change scenario turned into one interaction.
2. **Launch-window heatmap.** A grid of dates by sites, colored by sunlit % or Earth-access %, so the viable windows are visible at a glance. It becomes the signature screen of the app.
3. **Lander profile presets.** Let the user enter their own lander limits: how many hours of darkness it can survive, the minimum sun elevation for power, the comms elevation mask. Viability is then judged against those limits, not a generic score. Ship editable placeholders and don't claim real vehicle specs unless they are sourced.
4. **Stand-there view.** A first-person 360° panorama from the landing site: the terrain silhouette, the Sun and Earth discs moving as time is scrubbed, and sunset and Earthset countdowns. The horizon chart already holds the data; this makes it visceral for the public.
5. **Time-driven polar lighting.** Make the 2D hillshade follow the Sun's azimuth and elevation at the chosen time, so scrubbing the timeline visibly moves the shadows across Shackleton, Malapert, and the Connecting Ridge.
6. **An honesty layer.** Put a small badge on each metric: modeled or measured, data source, and resolution. Replace the binary sunlit/dark test with the fraction of the Sun's disc visible, since the Sun is about 0.5° wide and sunrise at a rim is gradual. Few tools show their own uncertainty.
7. **Earth-vs-Moon contrast.** Let a user pick a familiar city and compare its day length and temperature swing to the lunar site's, side by side. The challenge explicitly asks for the extreme differences to be obvious.
8. **Mission replay.** Jump to a real landing (Apollo 11 on 20 July 1969, IM-1 on 22 February 2024, Blue Ghost on 2 March 2025) and show the Sun and Earth geometry at touchdown. It is both teaching material and a validation test. Check that the ephemeris range covers 1969 first, and confirm each touchdown time against official sources.

9. **Plain-language summary.** Generate a deterministic sentence from the metrics, for example "This site is dark for 11.2 days starting 14 March, so a lander needs to survive a long night." Using rules instead of an AI model keeps it checkable.
10. **Handoff package.** A permalink, a one-page site card (PNG or PDF), a CSV of access windows, and an .ics file so comms windows land in a team calendar. This supports the challenge's science-to-engineering coordination goal.
11. **Three audiences, one app.** A Planner mode (dense numbers), an Educator mode (annotated charts), and a Public mode (guided tour, glossary).

9. A tight MVP definition

Promise: pick a landing site and date, see Sun and Earth against the real horizon, compare alternatives, and share the result.

The core loop is five steps:

1. Pick a site (catalog, search, or custom coordinates).
2. Pick a date and time.
3. See Sun and Earth against the horizon, with countdowns and access windows.
4. Compare with another site or another date.
5. Share a link or export a site card.

Cut or defer for now:

- **The Lunar & Space Weather modal.** The Moon has no weather, and the code comments name NOAA SWPC as the source; that solar-wind data is measured near Earth's L1 point, not at the Moon. Rename it "Thermal & space environment" or leave it out of the MVP.
- Auto-rotate, vector arrows, and the Earth 3D globe, unless they serve the loop.
- Mission Explorer polish; keep it as a simple picker.
- Contour-line and diagnostic controls in the 2D legend (the "1-Tile Gray" and "All Gray" modes are developer tools, not user features).

10. Three-minute demo script

Tell one story: **"Your launch just slipped."**

1. Open a south-pole site at the planned date. Show the verdict and its inputs.

2. Drag the date forward. The heatmap shows the window closing and the dark period jumping.
3. Click Plan B. The app suggests a later date at the same site and a nearby alternative.
4. Compare the two options with overlaid timelines.
5. Open the stand-there view and scrub to sunset. Show the Earthset countdown.
6. Share the permalink and download the site card.
7. Close on the validation page: your numbers against JPL Horizons.

11. Validation and credibility checklist

- [ ] Compare Sun and Earth elevation and azimuth against JPL Horizons at 5–10 site-times, and report the largest error
- [ ] Compare illumination % at Malapert and the Connecting Ridge against published LOLA-derived illumination maps
- [ ] Replay known landings and confirm the geometry looks right
- [ ] Unit-test edge cases: azimuth wrap at 0°/360°, a site at exactly 90°S where azimuth is undefined, dates at the ends of the supported range
- [ ] State limitations plainly: DEM resolution, no atmosphere refraction (none on the Moon), whether libration is included, how partial Sun-disc visibility is treated

12. Submission checklist

- [ ] Public repo with a README (setup, screenshots), a license, and no committed API tokens (check the Cesium ion token)
- [ ] Demo video under the official length limit (check the current Space Apps rules)
- [ ] Data credits: NASA LOLA (PDS), NASA Moon Trek, JPL Horizons if used, CesiumJS, three.js, Leaflet, d3
- [ ] Follow the Space Apps brand guidelines and don't use NASA insignia as your own logo
- [ ] Offline-safe demo: bundle textures locally, cache key tiles, and have a fallback if Moon Trek is down
- [ ] Fresh-machine test: clone, install, and run from a clean checkout